

Calabi Hats

Hi-Hat Engine · User Manual · v1.0

01 Overview

Calabi Hats is a hi-hat instrument built around three voiced layers, a sixteen-step arpeggiator with per-step probability, a Euclidean pattern generator and two modulation lanes you route by dragging them onto any knob.

It runs as VST3, Audio Unit, CLAP and as a standalone application, on macOS (universal, Apple Silicon and Intel) and on Windows 64-bit.

02 Installation

Format	macOS	Windows
VST3	~/Library/Audio/Plug-Ins/VST3	C:\Program Files\Common Files\VST3
Audio Unit	~/Library/Audio/Plug-Ins/Components	not available on Windows
CLAP	~/Library/Audio/Plug-Ins/CLAP	C:\Program Files\Common Files\CLAP
Standalone	run the application	run the executable

On macOS, files that arrive as a download carry a quarantine flag. Clear it once, in the Terminal, before opening your host:

```
xattr -dr com.apple.quarantine ~/Library/Audio/Plug-Ins/VST3/"Calabi Hats.vst3"
```

Then rescan your plugin folders in the host. Bitwig does not rescan on its own: go to Settings, Locations, and press rescan.

03 How a hit is built

Every hit is assembled from three layers — Metal, Click and Hiss — plus a short FM transient. Their sum passes through the comb filter, the high-pass filter, the ring modulator and the stereo spread, and the amplitude envelope shapes the result.

Modulation is resolved at the instant of each hit rather than continuously, so two consecutive steps can sound completely different without any smearing between them.

04 E-Hat Voice

- **Attack** — from an instant click to a soft swell. Below one millisecond for closed hats.
- **Decay** — the length of the hit. Short for closed, long for open.
- **Metal (M)** — six square oscillators tuned to the classic 808 ratios through a band-pass filter. Drag the M dot horizontally for frequency, vertically for level.

- **Click (C)** — a filtered noise burst that gives the attack its bite.
- **Hiss (H)** — high-passed noise, the air of the sound.
- **Comb Freq** — tunes the comb delay and places its metallic resonance.
- **Comb Feedback** — how long that resonance rings. High values turn the hat into a pitched metallic tone.
- **Comb Mix** — balance between the dry layers and the comb.
- **FM Hit Freq / Decay** — a short frequency-modulated transient. Raise the decay for a bell-like tail.
- **Filter / Q** — high-pass filter with resonance. Your main tool for sitting the hat above the kick.
- **Modulation Freq / Amount / Mix** — ring modulation. Low frequencies give tremolo, high frequencies give inharmonic metal.
- **Vel Sens.** — how strongly the per-step velocity changes the level.
- **Width** — stereo spread by delaying one channel. At zero the hat is mono.
- **Output** — the voice level.

05 Arpeggiator

- **ARP ON** — starts and stops the sequencer.
- **Direction** — Forward, Backward, Random, Odd steps or Even steps. Odd plays only the first, third, fifth step and so on; Even plays only the second, fourth, sixth.
- **Division** — step length, from a quarter note to a thirty-second, including triplets.
- **Octaves** — number of octaves the pattern climbs through before repeating.
- **Steps** — pattern length, from one to sixteen steps.
- **Gate** — scales the decay of every hit without touching the Decay knob.
- **Swing** — delays the even-numbered steps to push the groove.

The grid below has four rows. Tap a cell in ON to switch a step on or off, and drag vertically across PITCH, VEL and PROB to draw them.

- **ON** — whether the step plays at all.
- **PITCH** — transposition from minus twelve to plus twelve semitones, applied to the whole voice.
- **VEL** — step velocity, scaled by Vel Sens.
- **PROB** — the chance that the step actually sounds. At one hundred it always plays; lower values make the pattern breathe differently on every pass.

The Random notes button fills the whole grid at random: pitch, on and off, velocity and probability, with no musical constraint.

06 Euclidean Generator

A Euclidean pattern spreads a number of pulses as evenly as possible across a number of steps. It is how most traditional rhythms are built, and it is the fastest way to get a pattern that is neither square nor random.

- **Pulses** — how many hits to distribute.
- **Steps** — across how many steps. Generate also sets the arpeggiator length to this value.
- **Rotate** — shifts the whole pattern, which changes where the accent falls.

- **Accent** — how deep the difference is between a pulse and a gap.
- **Target** — what Generate writes: the on and off steps, the velocities, the probabilities, or accents, which leaves every step on and shapes velocity and probability instead.

The strip of dots above the readout previews the pattern before you commit it.

07 Mod Lanes and routing

Each lane is a sixteen-step curve you draw with the mouse. The two lanes run on their own clock divisions, so they drift against the arpeggiator and against each other.

- **Division** — how fast the lane advances.
- **Direction** — Forward, Reverse, Ping-Pong or Random.
- **Range** — Unipolar runs from zero upward; Bipolar runs from minus one to plus one, so the centre of the lane leaves the parameter untouched.
- **Amount** — the depth applied to every destination of that lane.
- **Tools** — Random, Ramp, Euclid, Reverse, Mirror, Half and Clear reshape the lane in one click.

To route a lane, drag its chip onto any knob. The outer ring shows the depth and direction of the modulation, and a coloured dot appears under the knob. Click that dot to remove the routing. One lane can drive as many knobs as you want.

Besides the two lanes, VEL sends the velocity of each step and RAND sends a new random value on every hit. Their depth is set by their own knobs in the Mod Sources panel.

08 MIDI and host sync

In a host the arpeggiator follows the transport and locks to the project position, so it always restarts in phase. In the standalone application it free-runs at the chosen tempo.

Any incoming MIDI note triggers a hit immediately. Middle C is the base pitch, and every semitone away from it transposes the metal, the click, the hiss and the FM transient together.

09 Presets and automation

Every knob and switch is exposed to the host, so it can be automated and recorded like any other parameter. The step grid, the two lanes and every modulation routing are stored with the host preset, so saving your project saves the whole pattern.

10 Troubleshooting

- **The host does not list the plugin** — check that the file sits in the folder in the installation table, then rescan. Bitwig needs the rescan button in Settings, Locations.
- **macOS refuses to open it** — clear the quarantine flag with the command in the installation section.
- **The Audio Unit is rejected** — run auval -v aumu CbHt CyAu in the Terminal. Apple's validator reports the exact reason the host used to reject it.
- **Ableton keeps it disabled** — Live remembers a plugin that failed once. Close Live and delete its plugin database folder, then start it again.

11 Specifications

Item	Detail
Formats	VST3, Audio Unit, CLAP, Standalone
macOS	universal binary, Apple Silicon and Intel, 10.13 or later
Windows	64-bit
Audio	stereo output
MIDI	note input
Sequencer	16 steps, per-step pitch, velocity and probability
Modulation	2 lanes plus velocity and random, free routing